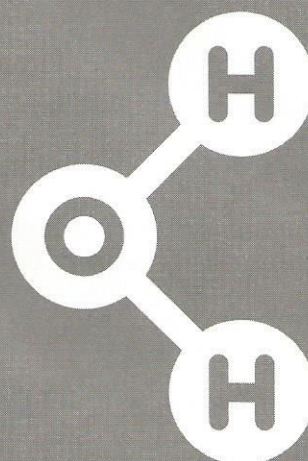




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C



2010
ICAS

International Competitions
and Assessments for Schools

SCIENCE

**Educational
Assessment
Australia**
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DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 55 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. The table shows the average heights of some animals at a zoo.

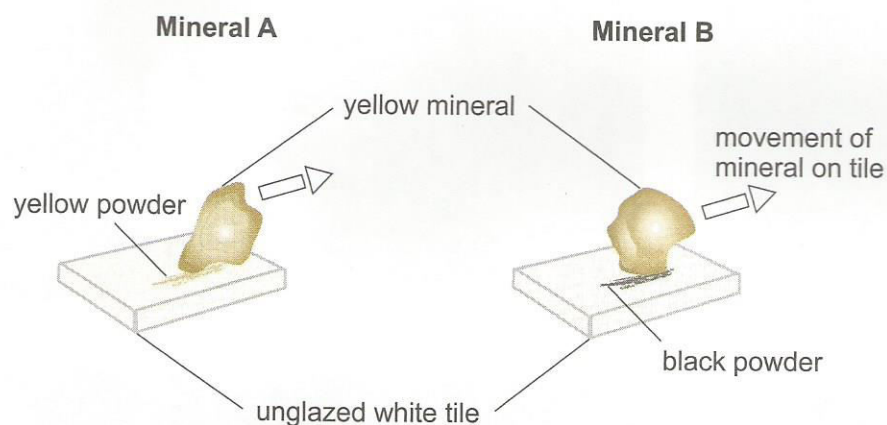
Animal	Average height
zebra	1.5 m
lemur	35 cm
hyena	80 cm
wildebeest	1.2 m

Which of these animals has the largest average height?

- (A) zebra (B) lemur (C) hyena (D) wildebeest

2. Minerals can be distinguished by their properties such as lustre (shine) and streak.

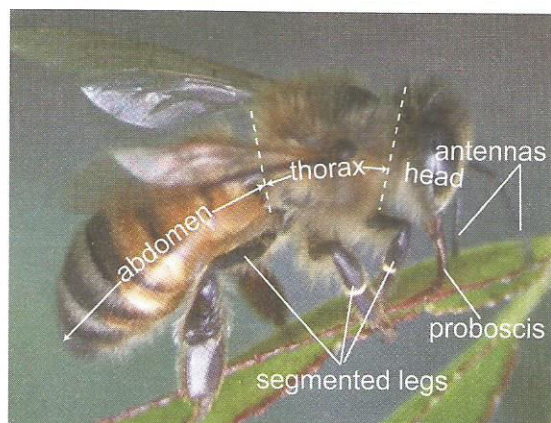
When a mineral is scratched on a rough surface that is harder than itself, a powder is left behind. The colour of the powder is called the streak of the mineral.



Using only the diagram, which property best distinguishes Mineral A from Mineral B?

- (A) lustre (B) colour (C) streak (D) hardness

3. The labelled photograph shows some body parts of a honey bee.



To which body part are the segmented legs attached?

- (A) abdomen
(B) head
(C) proboscis
(D) thorax

4. This is the nutrition information for a chocolate chip muesli bar.

SINGLE SERVE 31.3 g	QUANTITY PER SERVING	QUANTITY PER 100 g
ENERGY	540 kJ	1740 kJ
PROTEIN	2.1 g	6.8 g
FAT, TOTAL	4.1 g	13.1 g
- saturated	1.4 g	4.6 g
CARBOHYDRATE	20.1 g	64.2 g
- sugars	5.9 g	18.8 g
DIETARY FIBRE	2.0 g	6.3 g
SODIUM	80 mg	255 mg

Which of the following nutrients does a chocolate chip muesli bar have the most of?

- (A) Fat
- (B) Protein
- (C) Sodium
- (D) Carbohydrate

5. The table describes how different characteristics of a skateboard's wheel affect a skateboard's performance.

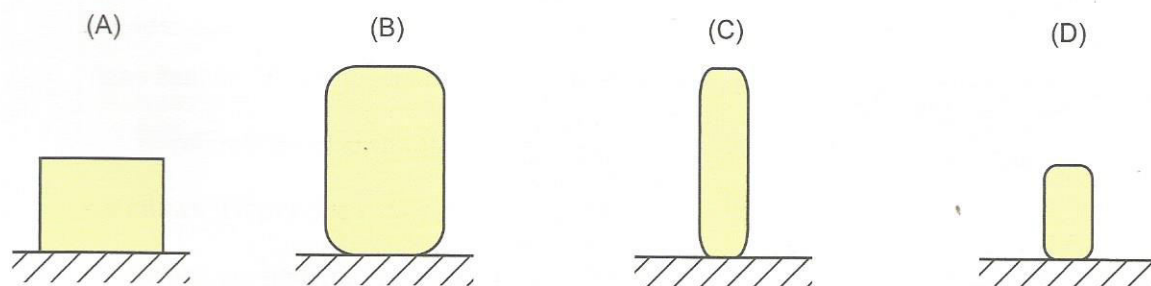
Wheel characteristic	Skateboard performance			
	Stability	Smooth ride	Fast ride	Tight turns
height (100 mm)	–	–	✓	X
height (50 mm)	–	–	✓	✓
wide	✓	✓	✓	X
narrow	X	X	✓	✓
flat surface on road	✓	✓	X	X
rounded surface on road	X	X	✓	✓

KEY

–	depends on other factors	✓	good performance	X	poor performance
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The scale diagram below shows the front views of four skateboard wheels.

Which wheel would be best suited for performing fast tight turns on a skateboard?



6. The photograph shows a moth at rest.

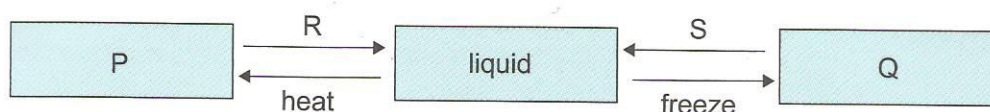


What is the moth's wingspan (mm) from X to Y?

- (A) 45 (B) 42 (C) 39 (D) 36

7. Matter can exist in three states; solid, liquid or gas, depending on temperature.

The diagram shows how matter can change from one state to another.



Which labels should be placed in the diagram?

	P	Q	R	S
(A)	gas	solid	heat	cool
(B)	gas	solid	cool	heat
(C)	solid	gas	heat	cool
(D)	solid	gas	cool	heat

8. To stay healthy we must exercise for about 30 minutes a day, 3 to 4 times a week. Thirty minutes of exercise can be one session for 30 minutes or many sessions of less time that add up to 30 minutes, for example 3 sessions of 10 minutes each.

The most beneficial exercise raises our heart rate by 50%.

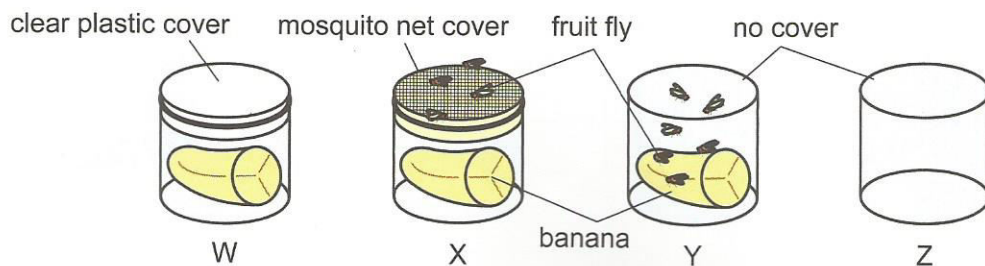
Ruby's heart rate when she is sitting quietly is 78 beats per minute. The table shows Ruby's heart rate during different activities.

Activity	Ruby's heart rate (beats per minute)
standing	85
walking	110
cycling	130
jogging	130
running	156
sprinting	180

From this information, which exercise will most benefit Ruby's health?

- (A) standing and talking to friends for 30 minutes each lunch time
 (B) walking to school for 40 minutes on Monday and Wednesdays
 (C) ten 60 second sprints with a short rest between each sprint, 4 times a week
 (D) jogging for 10 minutes, resting, then jogging again for 20 minutes, 3 times a week

9. Jane set up an investigation in her garden. She used four identical glass jars. The diagram shows how the setup looked after five hours.



What prediction about fruit flies would this experiment best test?

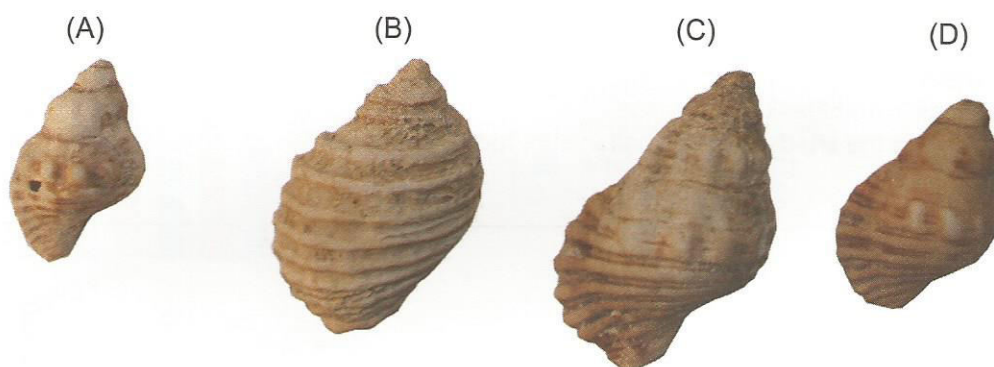
Fruit flies will be attracted to

- (A) open jars. (B) covered jars.
(C) the smell of bananas. (D) the colour of bananas.

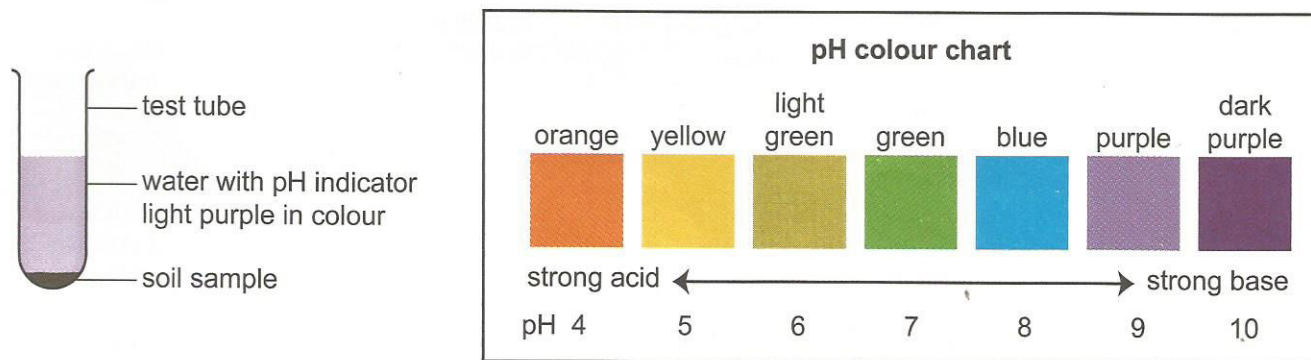
10. The photographs show four shells. The shells are all placed with the same orientation.

Three of the shells belong to the same group but are different sizes. One shell belongs to a different group.

Which shell most likely belongs to a different group?



11. Tim wanted to test the pH of some soil. He mixed the soil with water in a test tube and added five drops of indicator from his pH testing kit. He matched the colour produced in the test tube against the chart supplied with the kit.

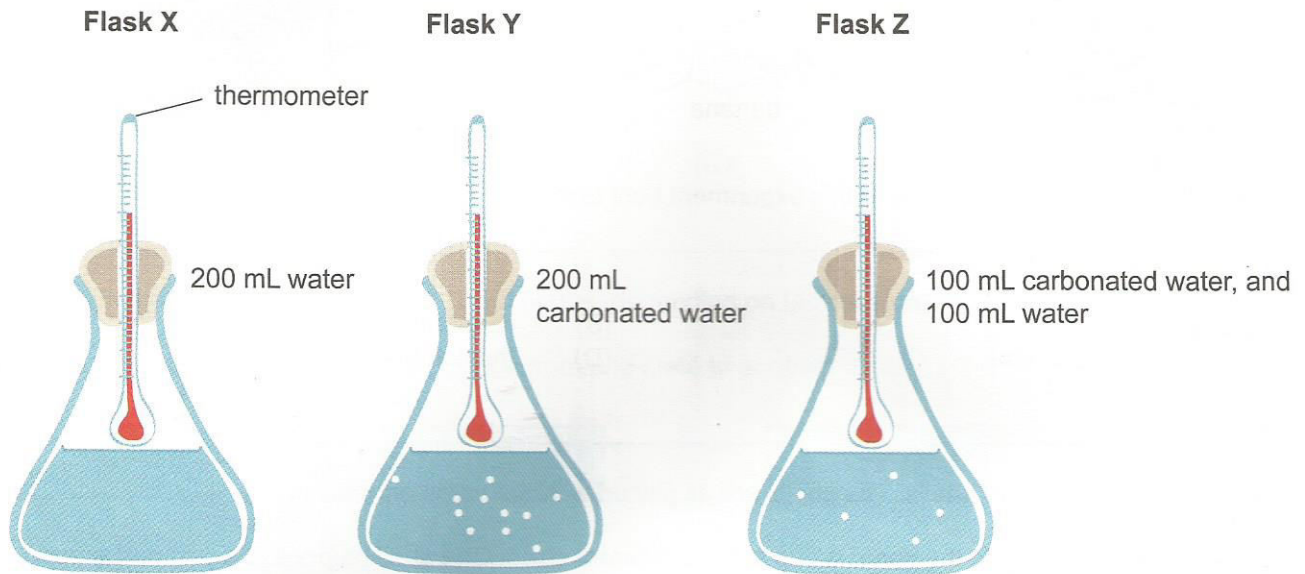


What is the approximate pH value of Tim's soil sample?

- (A) 4.5 (B) 6.5 (C) 8.5 (D) 10.5

12. More heat from the Sun is being trapped by rising levels of greenhouse gases, such as carbon dioxide, in the air. This causes an increase in the warming of the Earth.

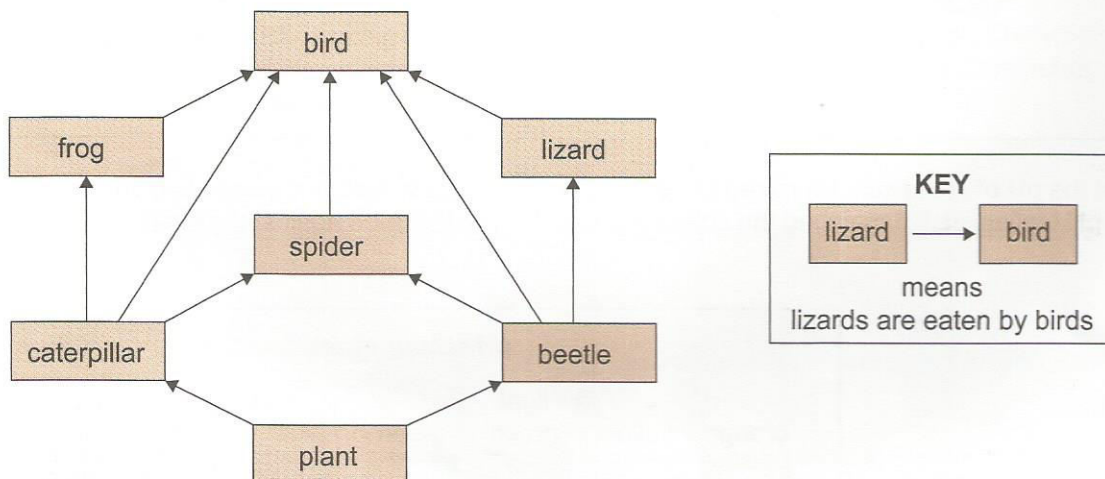
A group of students wanted to investigate the effect of carbon dioxide on air temperature. The diagram shows the setup at the start of their investigation. The carbonated water will release carbon dioxide into the air inside the flask.



Where should the students place the flasks?

- (A) all three flasks in the sun
- (B) all three flasks in the shade
- (C) Flask X in the shade, Flask Y and Flask Z in the sun
- (D) Flask X in the sun, Flask Y in the shade, Flask Z half in sun, half in shade

13. The diagram shows what some animals eat.



A disease killed all of the birds in this food web. Which of the following animal populations is most likely to decrease in number?

- (A) frog
- (B) lizard
- (C) beetle
- (D) spider

14. The table shows the highest temperature and how much rain fell each month.

Month	September (S)	October (O)	November (N)	December (D)	January (J)
Highest temperature (°C)	25	30	28	35	38
Rainfall (millimetres)	108	110	80	43	30

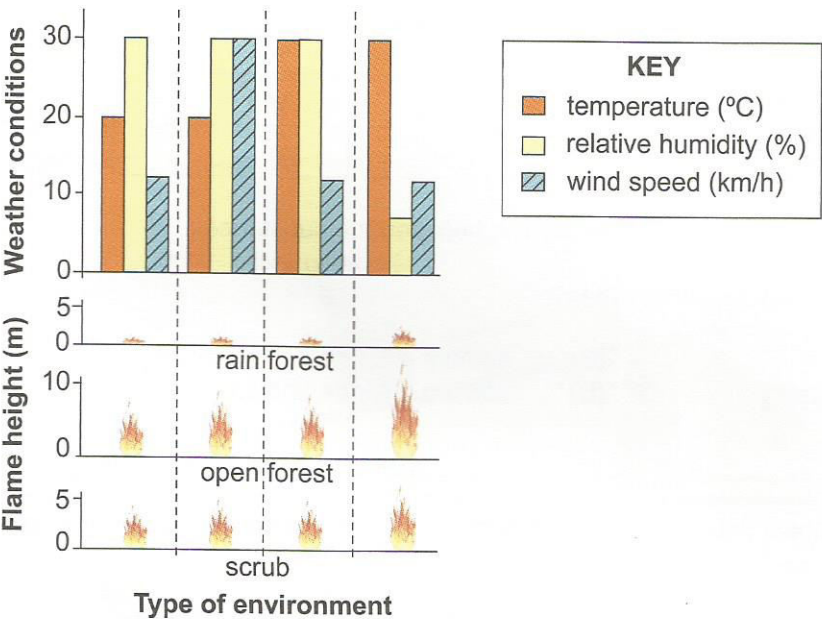
Which month had the most rainfall?

- (A) September
- (B) October
- (C) December
- (D) January

15. The type of environment effects the amount of fuel available to burn during a bushfire.

Relative humidity is the percentage of water vapour present in the air, compared to the maximum amount that the air can hold.

The graphs give information about the effects of four different sets of weather conditions on the flame height of bushfires in three different types of environments.



During a bushfire, which type of environment supports the highest flame, and which weather condition has most effect on the flame height?

	Environment supporting the highest flame	Weather condition most affecting flame height
(A)	open forest	temperature
(B)	open forest	relative humidity
(C)	scrub	relative humidity
(D)	scrub	temperature

16. Philip took some photographs of a flower over time.



Which option lists the photographs in the order in which they were taken?

- (A) X, W, Z, Y (B) X, Z, Y, W (C) W, X, Y, Z (D) W, Z, Y, X

17. Anne has some substances in her cupboard that have been used in cooking. They are stored in unlabelled plastic jars. The table gives some of the properties of the substances that are in the jars.

Substance	Looks	Tastes	Feels like	Dissolves in water?
sodium chloride	colourless	salty	small crystals	yes
calcium carbonate	white	sweet	fine powder	no
baking powder	white	sweet	small crystals	yes
sucrose	white	sweet	small crystals	yes
acetic acid	colourless	sour	water	yes

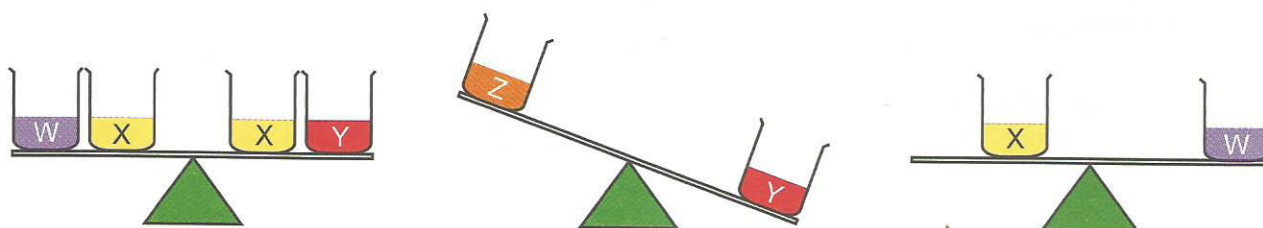
Using only the properties given in the table, Anne tried to distinguish between the substances.

Which two substances was she unable to distinguish between?

- (A) baking powder and sucrose (B) sodium chloride and acetic acid
(C) baking powder and calcium carbonate (D) sodium chloride and calcium carbonate

18. Liquids can be of different densities. A denser liquid will have a greater mass than the same volume of a less dense liquid.

The diagrams show a balance being used to compare the masses of equal volumes of four different liquids W, X, Y and Z.



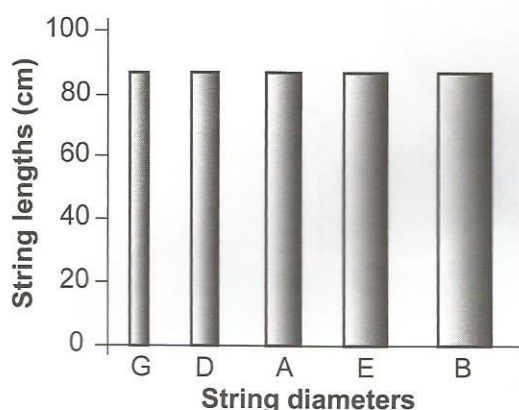
Which liquid is the most dense?

- (A) W (B) X (C) Y (D) Z

19. A guitar produces sound when its strings vibrate.

The number of times a string vibrates in one second is called its frequency. Frequency is measured in hertz (Hz). The greater the frequency, the higher the pitch of the sound produced.

The graph and table give information about the sound produced by a 5-string bass guitar.

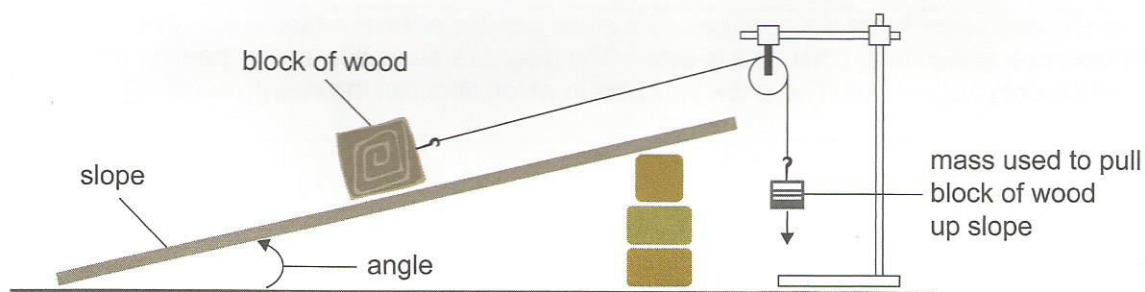


String	Frequency (Hz)
G	98
D	73.5
A	55
E	41
B	30.9

What can be inferred from this information?

- (A) Shorter strings produce a lower pitched sound.
- (B) Shorter strings produce a higher pitched sound.
- (C) Thicker strings produce a lower pitched sound.
- (D) Thicker strings produce a higher pitched sound.

20. Some students were studying friction. They set up the apparatus below to test how the surface and angle of a slope affect the movement of a block of wood.



In each trial, the students changed the angle of the slope and/or the surface, and adjusted the size of the mass until the block of wood just started to move up the slope.

The table shows the surface and angle of the slope in each trial.

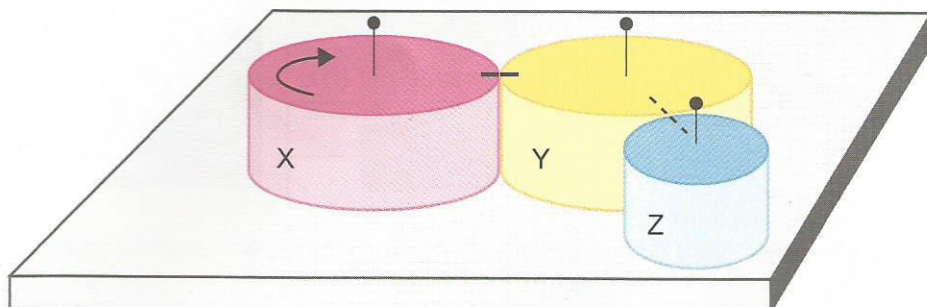
Trial	Surface	Angle
A	rough carpet	40°
B	rough carpet	30°
C	smooth polished wood	40°
D	smooth polished wood	30°

Which trial required the largest mass to pull the block of wood up the slope?

- (A) A
- (B) B
- (C) C
- (D) D

21. Fiona built a three-wheeled gear chain by connecting the gears by their outer edges (rims). She used two identically sized wheels and one wheel with half the diameter of the other two.

Fiona made a solid mark where wheel X touched wheel Y and a dotted mark where wheel Y touched wheel Z. She made sure that when she turned one wheel the other two wheels also turned without slipping.

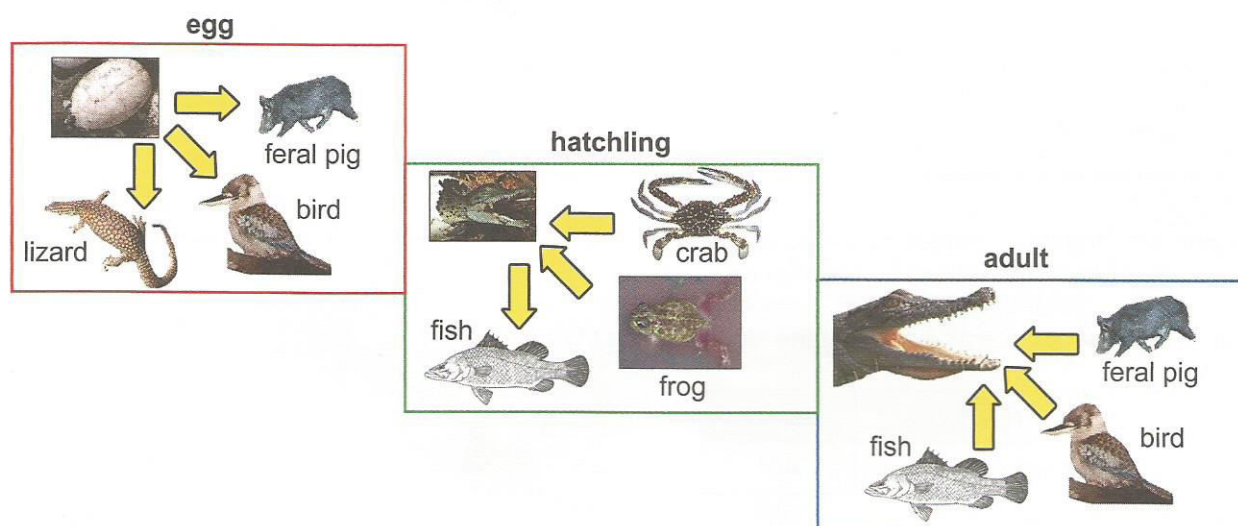


Fiona turned wheel X **two** times clockwise in the direction shown on the diagram.

How many times will wheel Z turn, and in what direction?

Wheel Z		
	Number of turns	Direction of motion
(A)	4	clockwise
(B)	1	clockwise
(C)	4	anticlockwise
(D)	1	anticlockwise

22. Energy transfers from one animal to another as it is eaten. The diagrams show this energy transfer for three stages of the life of *Crocodylus porosus*. The arrow indicates in which direction the energy transfers.

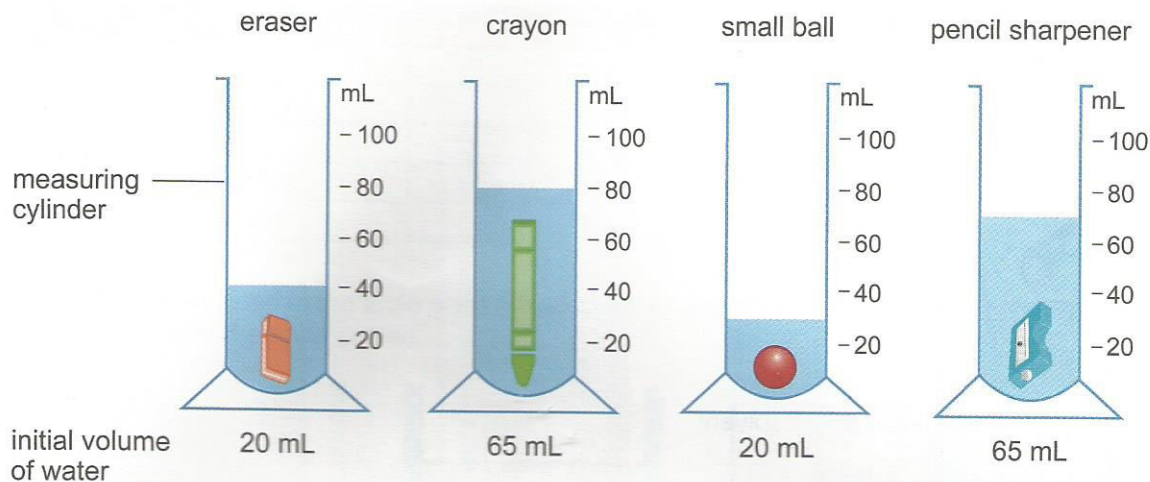


What eats *Crocodylus porosus* at an early stage, and then is eaten by *Crocodylus porosus* at a later stage?

- (A) lizards only
 (B) crabs and frogs only
 (C) fish, birds and feral pigs only
 (D) lizards, birds and feral pigs only

23. A student wanted to see which object had the greatest volume: an eraser, a crayon, a small ball or a pencil sharpener.

Initially a volume of water was placed in four identical empty measuring cylinders. An object was then carefully lowered into the water. The results are shown below.



Which object has the greatest volume?

- (A) eraser
- (B) crayon
- (C) small ball
- (D) pencil sharpener

24. In experiments, the position of a sample in relation to other samples should be random. This is called random design and ensures that the position of the samples does not affect the results of the experiment.

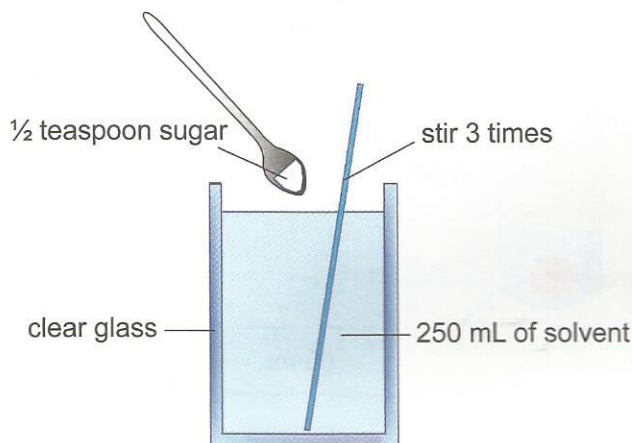
Which of the following is a random design?

- (A)
- (B)
- (C)
- (D)

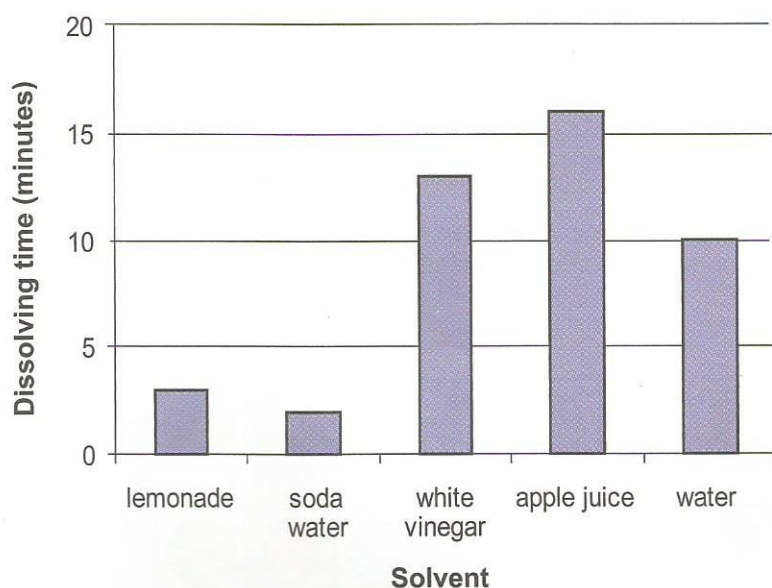
For questions 25 and 26 use the information below.

A solute is a substance (usually a solid) that gets dissolved. A solvent is a substance (usually a liquid) that does the dissolving. A solution is a mixture of a solute dissolved in a solvent.

Simon dissolved sugar in five different solvents: lemonade, soda water, white vinegar, apple juice and water. He predicted that the sugar would dissolve fastest in water. He used the apparatus and procedures shown in the diagram to carry out his tests.



The graph shows Simon's observations.



25. What could Simon conclude from his results?

- (A) He had predicted correctly.
- (B) Sugar dissolved fastest in soda water.
- (C) Water held more sugar in solution than either lemonade or soda water.
- (D) More sugar was dissolved in the apple juice than in any of the other solvents tested.

26. Soda water is made by dissolving the gas carbon dioxide in water. Which word describes the carbon dioxide?

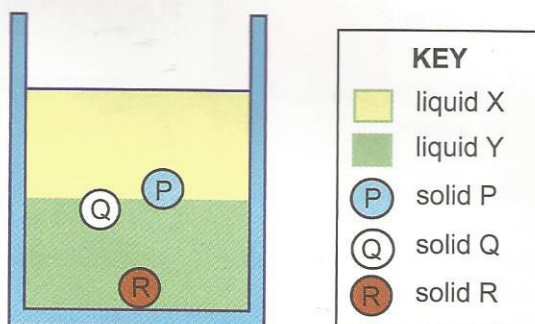
- (A) solute
- (B) solvent
- (C) mixture
- (D) water

27. After being mixed together, some liquids will separate again if left to settle. A less dense liquid will float on top of a denser liquid. A solid will sink in a liquid that is less dense than the solid, and float on a liquid that is more dense.

The table lists the density of some substances.

Substance	Density (g/cm ³)
detergent (liquid)	1.20
motor oil (liquid)	0.85
ice (solid)	0.92
ebony (solid)	1.10
PVC (solid)	1.40

The substances listed in the table were mixed together and then allowed to settle. The diagram shows the substances after they had settled.



What was liquid X and what was solid Q?

	Liquid X	Solid Q
(A)	detergent	ice
(B)	detergent	ebony
(C)	motor oil	ebony
(D)	motor oil	ice

28. Jason made four paper aeroplanes. The photograph shows Jason's aeroplanes.



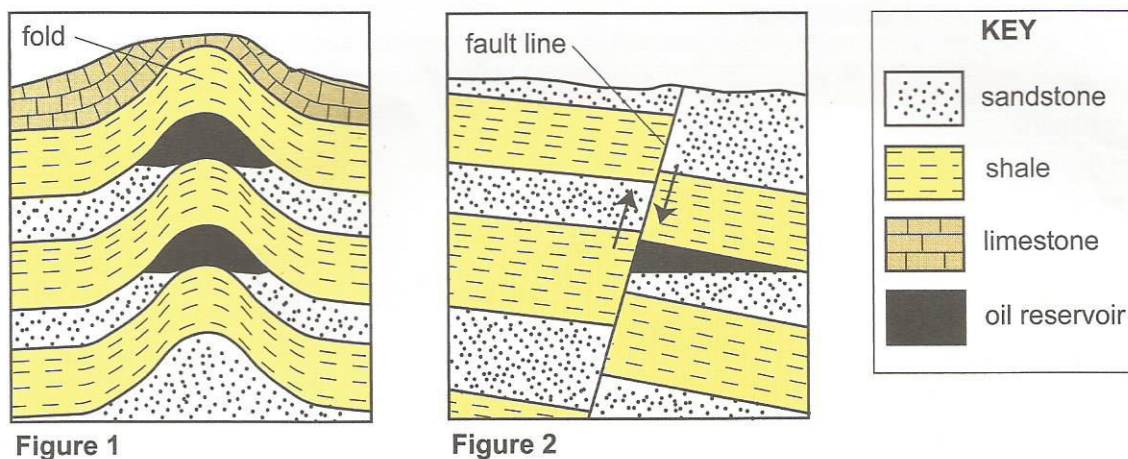
He wanted to find out which aeroplane will glide the furthest.

The results for which pair of aeroplanes will provide a fair test of how the folded winglets affect the distance an aeroplane will glide?

- (A) P and Q
- (B) Q and R
- (C) Q and S
- (D) R and S

29. Forces within the Earth

- fold rock layers as shown in Figure 1
- move rock layers upwards and downwards along a fault line as shown in Figure 2.



Permeable rocks allow fluids, such as gas, oil and water, to flow through them; impermeable rocks do not. Figure 1 and Figure 2 show where oil reservoirs may form.

For the oil reservoirs to form as shown above which conclusion can be drawn.

- (A) sandstone is permeable, limestone is impermeable
- (B) sandstone is permeable, shale is impermeable
- (C) shale is permeable, sandstone is impermeable
- (D) shale is permeable, limestone is impermeable

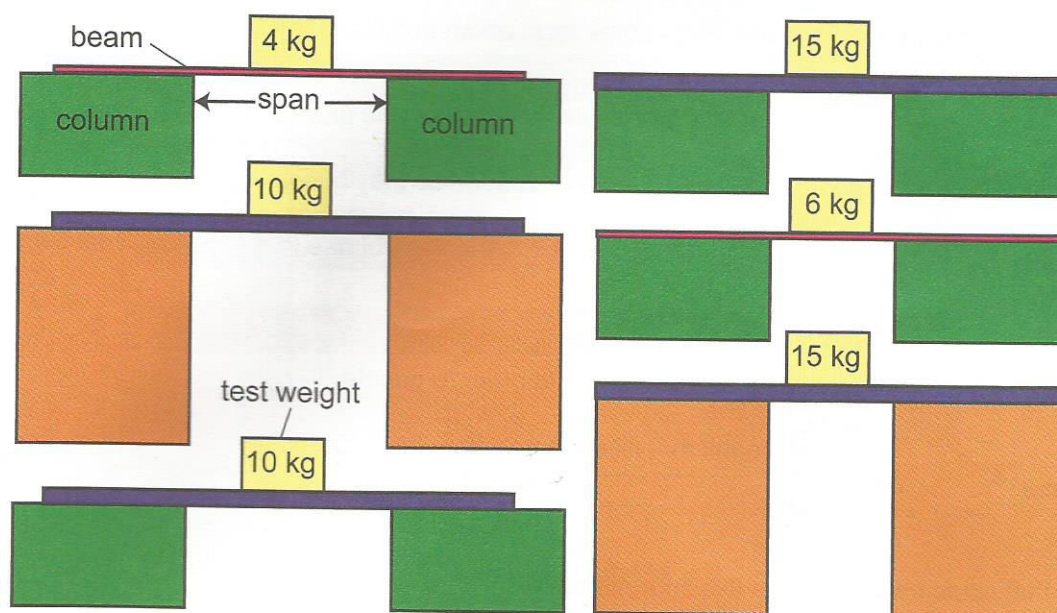
30. Lee keeps goldfish and catfish as pets. He wants to carry out a scientific investigation about his fish.

Choose the question that can be answered by carrying out a scientific investigation.

- (A) Are fish happy living in a fish bowl?
- (B) Are goldfish better pets than catfish?
- (C) Are goldfish more active in a bowl than in a tank?
- (D) Do catfish prefer to live in a tank alone or with other catfish?

31. Miriam built some bridges from columns and beams. She tested how strong the bridges were with weights.

The drawings show the bridges Miriam made, and the maximum weight each bridge beam could hold without breaking. The beams were made of the same material.



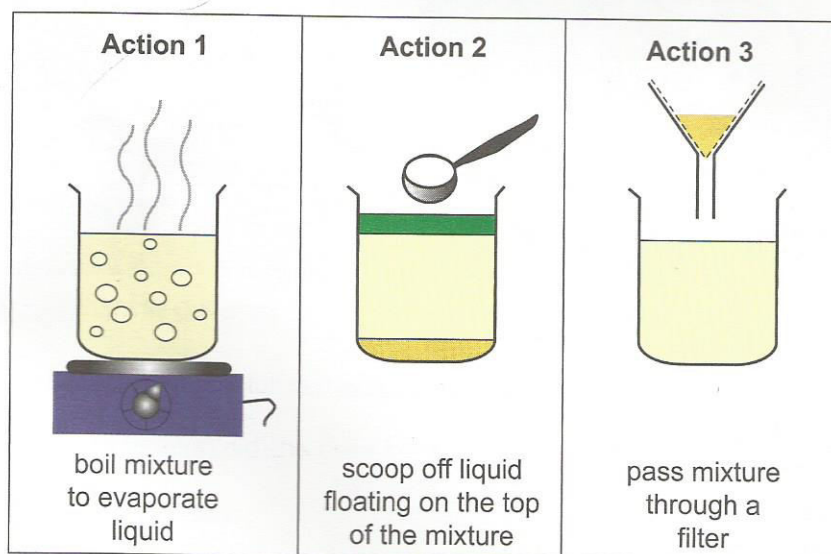
Based on Miriam's results, which two features of the bridges affected their strength?

- (A) beam thickness and column height
(B) beam length and column height
(C) beam thickness and span
(D) beam length and span

32. Here is some information about oil, water, sugar and sand.

- Oil floats on water.
- Sugar dissolves in water.
- Sand does not dissolve in water.

The diagrams show three actions taken to separate a mixture of oil, water, sugar and sand. The actions are not shown in the order they were performed.



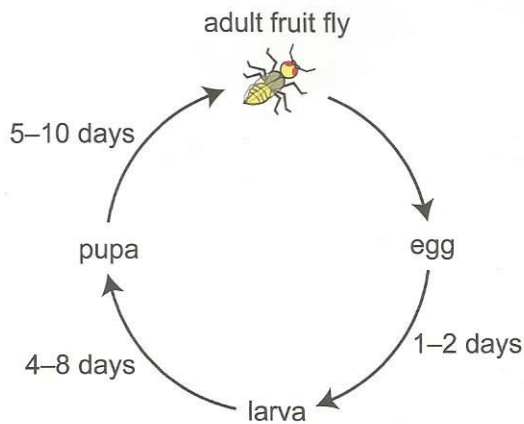
In which order were these actions performed to separate each part of the mixture from the others?

- (A) 2, 1, 3
(B) 2, 3, 1
(C) 3, 1, 2
(D) 3, 2, 1

For questions 33 and 34 use the information below.

The life cycle of an insect is the process of development from when an egg is laid to the time the insect is able to reproduce.

The diagram shows the life cycle of a fruit fly, and gives some facts about fruit flies.



adult fruit flies live 8–10 days and produce 400–500 eggs

fruit flies lay eggs as soon as they become adults

eggs are inserted into the flesh of ripe fruit by female fruit flies

larvae eat fruit

larvae leave fruit in order to pupate

adult fruit flies emerge from the pupae

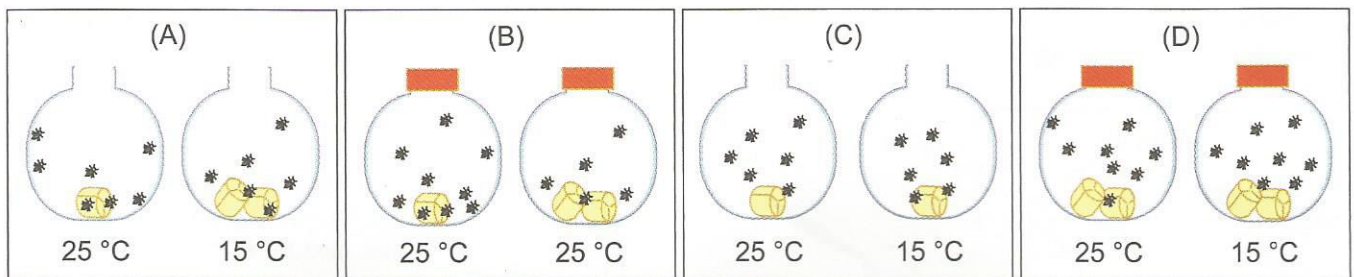
33. Which is a conclusion that could be drawn from this information?

Fruit flies

- (A) give birth to live young.
- (B) can reproduce in less than two weeks.
- (C) spend more than half their life eating fruit.
- (D) change to adults during the larval stage of their development.

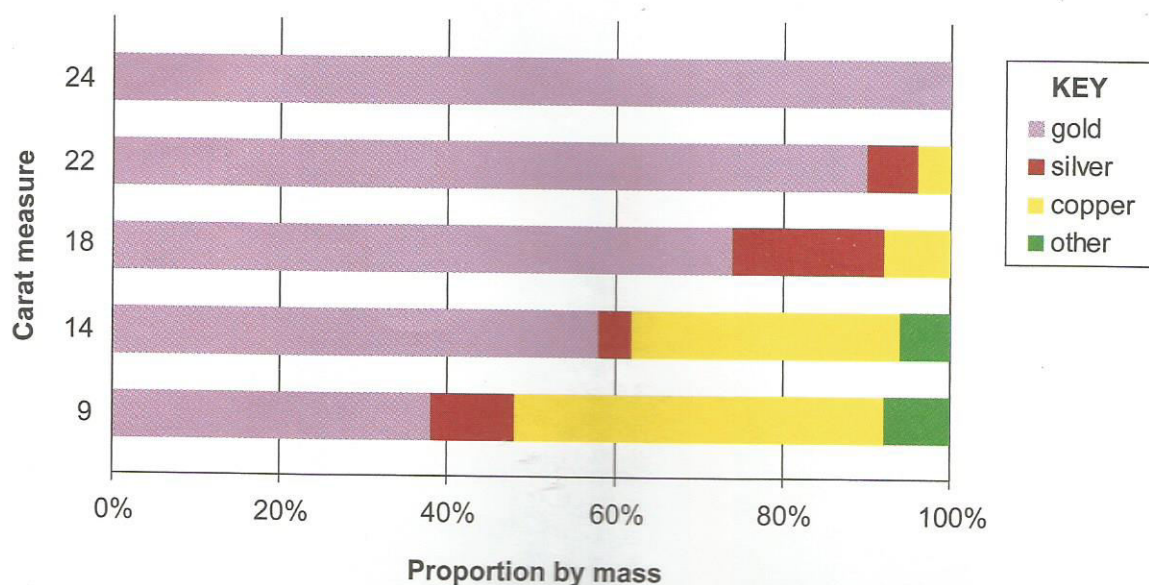
34. Simone read that temperature affected the length of the life cycle of fruit flies. She decided to investigate this idea. She trapped some fruit flies and placed them, together with pieces of banana, in glass jars. She observed the fruit flies for two weeks.

Which setup would give Simone a fair test of the effect of temperature on the life cycle of a fruit fly?



35. To increase the strength of pure gold when making jewellery, the gold is mixed with other metals, mainly silver and copper.

The graph shows mixtures of gold. The carat measure is the proportion by mass of pure gold in the mixture.



Approximately what mass of silver would be contained in a 50 gram ring made of 9 carat gold?

- (A) 5 g (B) 10 g (C) 24 g (D) 38 g

36. Lisa wanted to know whether a particular type of plant would grow best in a full sun or a shaded environment.

She planted four similar plants into four identical pots using the same type of potting mix, and watered each plant daily with the same amount of water. Lisa exposed each plant to a different light environment: from full sun to full shade. Her results after two weeks are shown in the photographs taken at the same scale.



Plant W (full sun)



Plant X (25% shade)



Plant Y (50% shade)



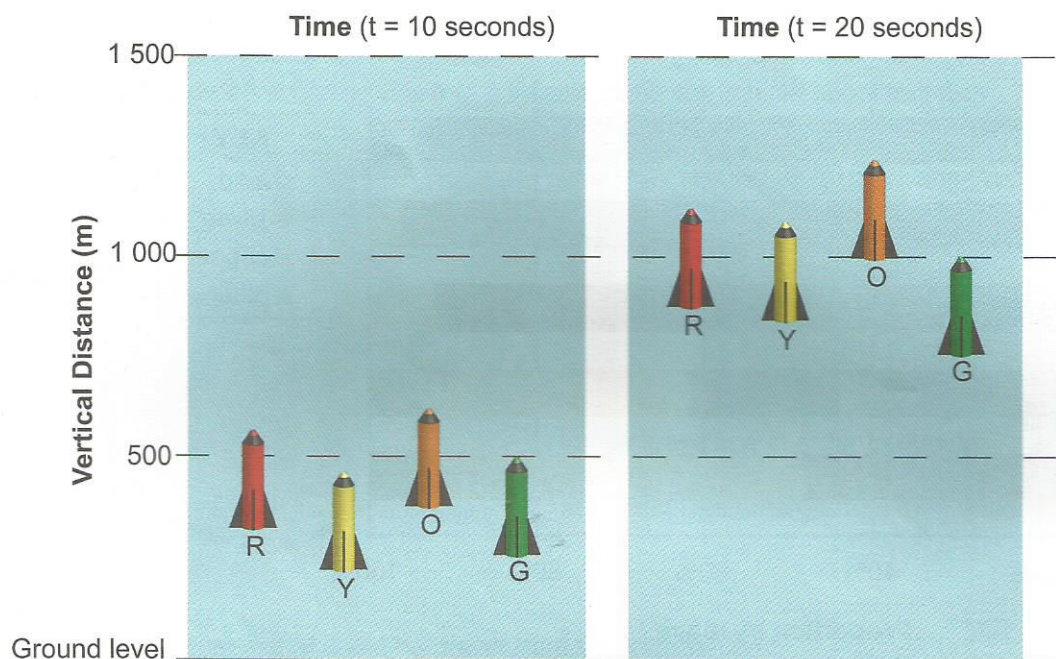
Plant Z (full shade)

Lisa concluded that full sun was the best light environment for growing this type of plant.

On which plant(s) did she base her conclusion?

- (A) Plant W only (B) Plant W and Z only
(C) Plant Z only (D) Plant W, X, Y and Z

37. The diagram shows the height (vertical distance) above ground of four rockets (R, Y, O and G) at 10 seconds and at 20 seconds after liftoff.



Speed is calculated by dividing distance travelled by the time taken. An object is accelerating if its speed changes over time.

Which rocket has the greatest upward acceleration between liftoff and 20 seconds after liftoff?

- (A) R (B) Y (C) O (D) G

38. The Earth and the Moon do not emit light. They reflect the light of the Sun.

Diagram 1 shows the illuminated shapes of the Moon (M) as seen from Earth (E) during a lunar month. The Moon waxes and then wanes during the lunar month.

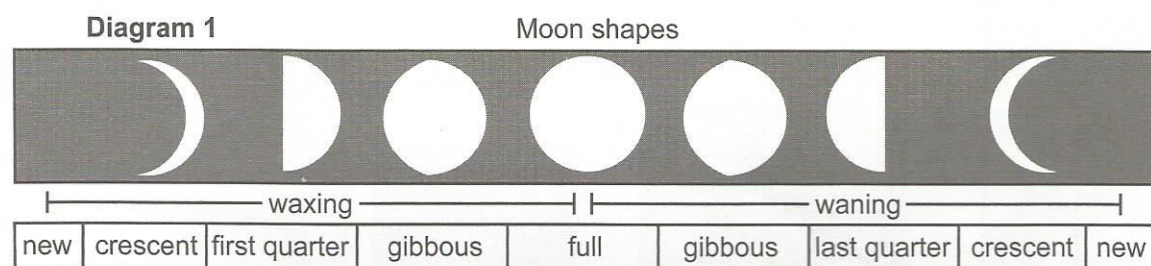


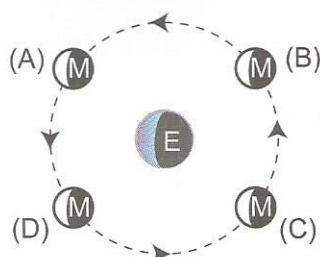
Diagram 2 shows the different positions of the Moon that produce some of the shapes shown in Diagram 1.

At which orbital position would the Moon be a waning crescent Moon?

Diagram 2

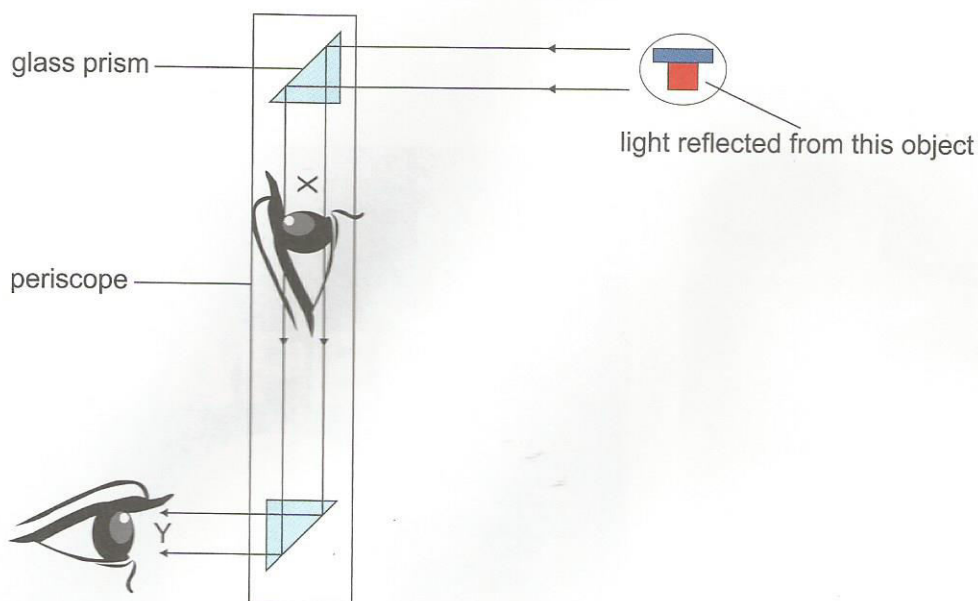


NOT TO SCALE



39. The rear surface of a glass triangular prism can reflect rays of light like a mirror. Glass prisms are used in a periscope to change the direction of light rays.

The drawing shows the path of light rays as they pass through a periscope.



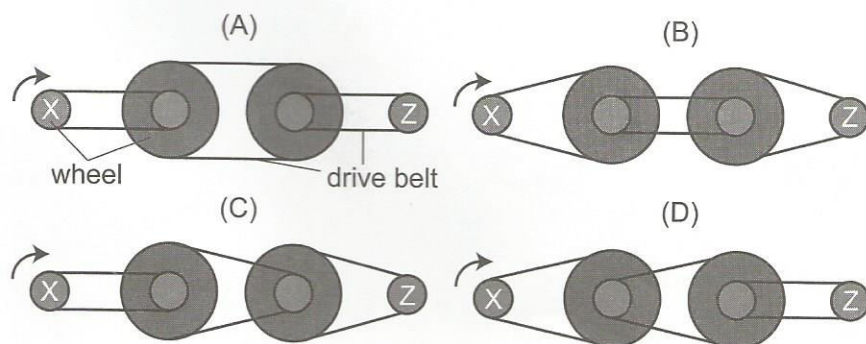
What would the image of the object look like at viewing point X, and what would the image of the object look like at viewing point Y?

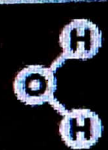
	Image at X	Image at Y
(A)		
(B)		
(C)		
(D)		

40. The diagram shows four different wheel systems that are driven by drive belts.

In each system, wheel X is rotated in the same direction at the same speed. Movement transfers through the system from wheel to wheel via the drive belts.

In which system would wheel Z move fastest?





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2010 Science Answer Keys

ICAS

International Competition in Academic Science
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
1	C	C	A	A	D	C	B	D	A
2	B	A	C	C	C	C	B	A	C
3	B	D	D	A	C	C	A	A	D
4	C	C	D	C	B	B	D	D	A
5	C	B	D	C	B	A	B	C	C
6	B	D	A	D	B	B	A	D	A
7	B	D	B	D	C	A	D	B	C
8	A	D	D	C	B	C	C	D	D
9	D	B	C	B	B	C	C	B	D
10	C	B	B	C	D	D	B	C	B
11	A	C	C	B	C	C	C	A	D
12	A	C	A	D	C	C	C	C	C
13	B	C	C	B	D	D	A	C	D
14	D	A	B	A	A	B	C	B	B
15	D	D	B	A	C	C	B	C	C
16	D	C	B	B	D	A	D	B	C
17	A	A	A	B	A	B	D	D	C
18	C	A	B	D	C	D	A	D	A
19	A	C	C	B	C	B	B	C	C
20	B	B	A	A	A	B	A	A	C
21	D	A	A	B	D	D	A	A	B
22	B	C	C	B	A	A	A	A	C
23	D	C	A	D	A	D	C	B	A
24	A	D	B	A	B	B	D	A	D
25	A	B	B	D	C	A	C	D	A
26	D	D	A	C	C	B	B	A	A
27	C	A	C	C	A	C	C	A	B
28	A	A	C	D	C	B	B	B	B
29	D	D	B	C	B	C	B	D	D
30	C	B	C	B	D	A	B	A	B

(Please turn over)



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2010 Science Answer Keys

ICAS
International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
31			C	B	B	B	B	D	A
32			B	C	B	D	D	A	A
33			B	D	B	B	B	C	D
34			D	B	D	B	D	B	D
35			A	D	B	D	C	B	C
36			D	D	A	D	B	C	C
37			B	A	A	D	D	A	B
38			A	B	C	C	A	D	A
39			D	A	B	D	D	B	D
40			C	A	D	D	D	C	C
41					A	A	B	C	B
42					D	A	A	B	A
43					D	A	B	B	B
44					D	B	C	C	B
45					B	B	C	C	A